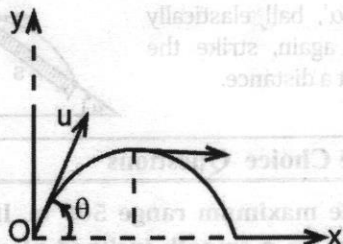


Chapter-5

PROJECTILE MOTION

- In projectile motion acceleration is due to gravity.
- The path of the projectile is called trajectory.
- Nature of trajectory is parabolic.
- Horizontal component of velocity is constant throughout the motion.



$$\text{Range, } R = \frac{u^2 \sin 2\theta}{g}$$

$$\text{Max}^m. \text{ Height, } H = \frac{u^2 \sin^2 \theta}{2g}$$

$$\text{Time of flight, } t_f = \frac{2u \sin \theta}{g}$$

⇒ Time of Ascent and time of descent are same.

$$t_a = t_d = \frac{1}{2} t_f = \frac{u \sin \theta}{g}$$

$$\frac{H}{R} = \frac{\tan \theta}{4}$$

$$\Rightarrow \text{For maximum range, } \theta = 45^\circ, R_{\max} = \frac{u^2}{g}, H = \frac{R_{\max}}{4}$$

- If a projectile is projected so that its range obtained is max^m, then max^m height attained by it is $\frac{1}{4}$ of maximum range.
- If a person can throw maximum horizontal distance R_0 , then he can throw maximum height $\frac{R_0}{2}$.
- Height is maximum if $\theta = 90^\circ$, $H_{\max} = \frac{u^2}{2g} = \frac{1}{2} R_{\max}$
- At highest point angle between velocity and acceleration is 90° .
- Velocity is min^m at highest point hence kinetic energy is minimum.

- If a projectile is projected with kinetic energy 'K' at an angle θ with horizontal then,

(i) KE at highest point or Min^m KE.

$$K_H \text{ or } K_{\min} = K \cos^2 \theta$$

$$\text{If } \theta = 30^\circ, K_H = \frac{3}{4} K = 75\% \text{ of KE}$$

(ii) Max^m potential energy or PE at highest point
 $U_{\max} \text{ or } U_H = K \sin^2 \theta$

$$\text{If } \theta = 30^\circ, U_H = \frac{1}{4} K = 25\% \text{ of KE}$$

- If a projectile is projected with linear momentum P at an angle ' θ ' with horizontal then change in linear momentum at any time.

$$\Delta p = mgt$$

i) If highest point $t = t_a = \frac{u \sin \theta}{g}$: $\Delta p = p \sin \theta$

ii) If striking point $t = t_f = \frac{2u \sin \theta}{g}$: $\Delta p = 2p \sin \theta$

e.g. i) if $\theta = 30^\circ$, Δp when projectile strike

$$\Delta p = p, \Delta p = 100\% \text{ of } p$$

- If a projectile is thrown with speed u at angle θ with horizontal the projectile makes an angle ' α ' with horizontal then its speed, $v = u \cos \theta \cdot \sec \alpha$
- Angle made by velocity after time 't' with horizontal and its value,

$$v = \sqrt{v_x^2 + v_y^2}$$

$$\tan \alpha = \left(\frac{u \sin \theta - gt}{u \cos \theta} \right) = \frac{v_y}{v_x}$$

- A projectile is projected with velocity 'u' at angle ' θ ' with horizontal then at time 't' its velocity v is at perpendicular to initial velocity then,

$$v = u \cot \theta, \quad t = \frac{u}{g} \operatorname{cosec} \theta$$

Eqⁿ of trajectory,

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

$$\therefore y = ax - bx^2$$

In terms of range and max^m height

$$(i) \quad y = \tan \theta \cdot x - \frac{\tan \theta}{R} x^2, \quad R = \frac{a}{b}$$

$$(ii) \quad y = \tan \theta \cdot x - \frac{\tan^2 \theta}{4H} x^2, \quad H = \frac{a^2}{4b}$$

- Average velocity during time of ascent (i.e., avg. velocity when projectile is at highest point)

$$V_{\text{avg}} = \frac{u}{2} \sqrt{1 + 3 \cos^2 \theta}$$

- Avg. velocity when projectile strikes to ground,

$$V_{\text{avg}} = u \cos \theta$$

- Change in speed when projectile is at highest point,

$$\Delta |\vec{v}| = u(1 - \cos\theta) = 2u \sin^2\theta/2$$

- If two projectile are projected with same speed at different angles then for same range,

- (i) Sum of angles of projection must be 90°

$$\theta + \alpha = 90^\circ,$$

(ii) $\frac{H_\theta}{H_\alpha} = \tan^2\theta$ or $\cot^2\alpha$

(iii) $\frac{(t_f)_\theta}{(t_f)_\alpha} = \tan\theta$ or $\cot\alpha$

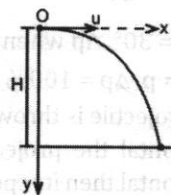
(iv) $R = \frac{1}{2}g(t_f)_\theta(t_f)_\alpha \Rightarrow R = \frac{1}{2}gt_1t_2 : t_1t_2 \propto R$

(v) $R = 4\sqrt{H_\theta H_\alpha}$
 $\Rightarrow R = 4\sqrt{H_1 H_2} \quad H_1 H_2 \propto R^2$

(vi) $H_1 + H_2 = \frac{u^2}{2g}$

Horizontal projection from a height

(i) $t_f = \sqrt{\frac{2H}{g}}$ = time of flight.



(ii) Velocity after time 't', $v = \sqrt{u^2 + g^2t^2}$

(iii) Angle made by velocity after time 't', $\tan\alpha = \frac{gt}{u}$ from horizontal

(iv) Range, $R = u \times \sqrt{\frac{2H}{g}}$

- When a ball rolled off from top of staircase with horizontal velocity 'u' having width 'b' and height 'h' the ball hits n^{th} step then,

$$n = \frac{2h u^2}{g b^2}$$

- If a man hits the target, he should point his gun in a direction higher than the target.
- If a man fires his gun directly aimed towards monkey at height, at same instant monkey starts falling then **bullet hits the monkey**.
- If two bodies are projected horizontally from certain height with different velocities u_1 and u_2 in opposite directions then.

- (i) their velocities are perpendicular after time,

$$t = \frac{\sqrt{u_1 u_2}}{g}$$

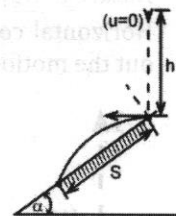
- (ii) velocities of 1st body and 2nd body when their velocity are perpendicular, $v_1 = \sqrt{u_1^2 + u_1 u_2}$ and $v_2 = \sqrt{u_2^2 + u_1 u_2}$

- (iii) their position vectors are perpendicular of after time,

$$t = \frac{2\sqrt{u_1 u_2}}{g}$$

- If a ball is dropped from height 'h' from the top of an inclined plane of inclination ' α ', ball elastically collides the it again, strike the inclined plane at a distance.

$$S = 8h \sin\alpha$$



Multiple Choice Questions

1. A projectile has the maximum range 500 m. If the projectile is thrown up a smooth inclined plane of 30° with the same (magnitude) velocity, the distance covered by it along the inclined plane till it stops will be:

- a. 250 m b. 500 m
c. 750 m d. 1000 m

2. From the top of a tower of height 40 m, a ball is projected upwards with a speed of 20 m/s at an angle of elevation of 30° . The ratio of the total time taken by the ball to hit the ground to its time of flight (time taken to come back to the same elevation) is (take $g = 10 \text{ m/s}^2$):

- a. 2 : 1 b. 3 : 1
c. 3 : 2 d. 1.5 : 1

3. The maximum height attained by a projectile is increased by 10% by increasing its speed of projection, without changing the angle of projection. The percentage increase in the horizontal range will be:

- a. 20% b. 15%
c. 10% d. 5%

4. A body is projected horizontally from a point above the ground. The motion of the body is described by the equations $x = 2t$ and $y = 5t^2$ where x and y are the horizontal and vertical displacements (in m) respectively at time t. The trajectory of the body is:

- a. a straight line b. a circle
c. an ellipse d. a parabola

5. A ball is projected with a velocity $20\sqrt{3}$ m/s at angle 60° to the horizontal. The time interval after which the vertical vector will make an angle 30° to the horizontal is (Take $g = 10 \text{ m/s}^2$):

a. 4 sec b. 2 sec
c. 1 sec d. 3 sec

6. A projectile is thrown upward with a velocity v_0 at an angle α to the horizontal. The change in velocity of the projectile when it strikes the same horizontal plane is:

a. $v_0 \sin \alpha$ vertically downward
b. $2v_0 \sin \alpha$ vertically downward
c. $2v_0 \sin \alpha$ vertically upward
d. $v_0 \cos \alpha$ vertically downward

7. The velocity of a projectile at any instant is u making an angle α to the horizon. The time after which it will be moving at right angle to this direction is:

a. $\frac{u \operatorname{cosec} \alpha}{g}$ b. $\frac{2u \sin \alpha}{g}$
c. $\frac{u \cos \alpha}{g}$ d. $\frac{u \tan \alpha}{g}$

8. The velocity at the maximum height of a projectile is half of its initial velocity of projection (u). Its range on horizontal plane is:

a. $\frac{3u^2}{g}$ b. $\frac{3}{2} \cdot \frac{u^2}{g}$
c. $\frac{u^2}{3g}$ d. $\frac{\sqrt{3}}{2} \cdot \frac{u^2}{g}$

9. A stone is projected in air. Its time of flight is 3 seconds and range is 150 m. Maximum height reached by the stone is ($g = 10 \text{ ms}^{-2}$):

a. 37.5 m b. 22.5 m
c. 90 m d. 11.25 m

10. A projectile A is thrown at an angle 30° to the horizontal from point P. At the same time another projectile B is thrown with velocity V_2 upwards from the point Q vertically below the highest point A would reach. For B to collide with A the ratio

$\frac{V_2}{V_1}$ should be:

a. 1 b. 2
c. $1/2$ d. 4

11. The equation of motion of a projectile is:

$$y = 12x - \frac{3}{4}x^2$$

Given that $g = 10 \text{ ms}^{-2}$, what is the range of the projectile?

a. 12 m b. 16 m
c. 20 m d. 24 m

12. A ball is thrown at different angles with the same speed u and from the same point. It has the same range in both the cases. If y_1 and y_2 be the heights attained in the two cases, then $y_1 + y_2 = \dots$:

a. $\frac{u^2}{g}$ b. $\frac{2u^2}{g}$
c. $\frac{u^2}{2g}$ d. $\frac{u^2}{4g}$

13. A ball is projected from the ground at angle θ with the horizontal. After 1 second it is moving at angle 45° with the horizontal and after 2 seconds it is moving horizontally. What is the velocity of projection of the ball?

a. $10\sqrt{3} \text{ ms}^{-1}$ b. $20\sqrt{3} \text{ ms}^{-1}$
c. $10\sqrt{5} \text{ ms}^{-1}$ d. $20\sqrt{2} \text{ ms}^{-1}$

14. A boy throws a ball with a velocity u at an angle θ with the horizontal. At the same instant he starts running with uniform velocity to catch the ball before it hits the ground. To achieve this he should run with a velocity of:

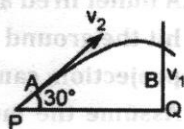
a. $u \cos \theta$ b. $u \sin \theta$
c. $u \tan \theta$ d. $\sqrt{u^2 \tan \theta}$

15. One second after the projection, a stone moves at an angle of 45° with the horizon. Two seconds from the start, it is travelling horizontally. Its angle of projection with the horizon is (given $g = 10 \text{ ms}^{-2}$):

a. 60° b. $\tan^{-1}(4)$
c. $\tan^{-1}(3)$ d. $\tan^{-1}(2)$

16. A projectile is thrown with an initial velocity of $(a\hat{i} + b\hat{j}) \text{ ms}^{-1}$. If the range of the projectile is twice the maximum height reached by it, then:

a. $a = 2b$ b. $b = a$
c. $b = 2a$ d. $b = 4a$



17. The equation of a projectile is $Y = \sqrt{3} X - \frac{1}{2} g X^2$.

The velocity of projection is:

- a. 1 m/s b. 2 m/s
c. 3 m/s d. 4 m/s

18. An object of mass m is projected with a momentum P at such an angle that its maximum height (H) is $\frac{1}{4}$ th of its horizontal range (R). Its minimum kinetic energy in its path will be;

- a. $\frac{P^2}{8m}$ b. $\frac{P^2}{4m}$
c. $\frac{3P^2}{4m}$ d. $\frac{P^2}{m}$

19. Two projectiles A and B are projected with same speed at angles 30° and 60° to the horizontal then, which one is wrong:

- a. $R_A = R_B$ b. $H_B = 3H_A$
c. $T_B = \sqrt{3} T_A$ d. none of these

(Here, R stands for range, H for maximum height and T for time of flight).

20. A ball of mass m is projected from the ground with an initial velocity u making an angle of θ with the horizontal. What is the change in velocity between the point of projection and the highest point:

- a. $u \cos^2 \theta$ downward
b. $u \cos \theta$ upward
c. $u \sin^2 \theta$ upward
d. $u \sin \theta$ downward

21. A ball, whose kinetic energy is E , is thrown at an angle of 45° with the horizontal, its kinetic energy at the highest point of its flight will be:

- a. zero b. $E/2$
c. $E/\sqrt{2}$ d. E

22. A solid sphere and a hollow sphere are thrown horizontally from a cliff with equal velocities respectively. Then which sphere reaches first on earth?

- a. solid sphere
b. hollow sphere

- c. both spheres simultaneously
d. we can not say because masses of spheres are not given

23. A stone is projected at an angle θ with the horizontal with velocity u . It executes a nearly circular motion near its maximum height for a short time. The radius of circular path is:

- (a) $\frac{u^2 \cos^2 \theta}{2g}$ b. $\frac{u^2 \sin^2 \theta}{g}$
c. $\frac{u^2 \cos^2 \theta}{g}$ d. $\frac{u^2}{g}$

24. A particle is projected at an angle of 45° from a point lying 2m from the foot of a wall. It just touches the top of the wall and falls on the ground 4 m from it. The height of the wall is:

- a. $\frac{1}{3}$ m b. $\frac{4}{3}$ m
c. $\frac{2}{3}$ m d. $\frac{3}{4}$ m

25. Two projectiles are projected with the same velocity. If one is projected at an angle of 30° and the other at 60° to the horizontal. The ratio of maximum heights reached is:

- a. 1 : 3 b. 2 : 1
c. 3 : 1 d. 1 : 4

26. A ball is thrown upwards at an angle of 60° to the horizontal. It falls on the ground at a distance of 90 m. If the ball is thrown with same initial velocity at an angle of 30° . IT will fall on the ground at a distance of:

- a. 120 m b. 90 m
c. 60 m d. 30 m

27. A bullet fired at an angle of 30° with the horizontal hit the ground 3 km away. By adjusting its angle of projection, can one hope to hits target 5 km away. Assume the muzzle speed of be same and the air resistance is negligible:

- a. possible to hit a target 5 km away
b. not possible to hit a target 5 km away
c. prediction is not possible
d. none of the above

28. For a given velocity, a projectile has the same range R for two angles of projection. If t_1 and t_2 are the time of flight in the two cases, then:

- a. $t_1 t_2 \propto R$ b. $t_1 t_2 \propto R^2$
 c. $t_1 t_2 \propto \frac{1}{R^2}$ d. $t_1 t_2 \propto \frac{1}{R}$

29. A bomber plane is moving horizontally with a speed of 500 m/s and a bomb released from it, strikes the ground in 10 sec. Angle from horizontal at which the bomb strikes the ground is:

($g = 10 \text{ m/s}^2$)

- a. $\tan^{-1}(1)$ b. $\tan^{-1}(5)$
 c. $\tan^{-1}\left(\frac{1}{5}\right)$ d. $\sin^{-1}\left(\frac{1}{5}\right)$

Answers:

1. b	2. a	3. c	4. d	5. b
6. b	7. a	8. d	9. d	10. c
11. b	12. c	13. c	14. a	15. d
16. c	17. b	18. b	19. d	20. d
21. b	22. c	23. c	24. b	25. a
26. b	27. d	28. a	29. c	

Solution for Selected Problems

1. $R_{mg} = \frac{u^2}{g} \Rightarrow u^2 = 500 \times 10 = 5000$

Distance covered by body along inclined plane before coming to rest

$$s = \frac{u^2}{2g \sin \theta} = \frac{5000}{2 \times 10 \times \sin 30^\circ} = 500 \text{ m}$$

2. Time taken to come back to same elevation

$$= \frac{2u \sin \theta}{g} = \frac{2 \times 20 \times \sin 30^\circ}{10} = 2 \text{ sec}$$

Total time take to hit the ground

$$= \frac{u \sin \theta}{g} + \sqrt{\left(\frac{u \sin \theta}{g}\right)^2 + \frac{2h}{g}}$$

$$= \frac{20 \times \sin 30^\circ}{10} + \sqrt{\left(\frac{20 \times \sin 30^\circ}{10}\right)^2 + \frac{2 \times 40}{10}}$$

$$= 1 + \sqrt{1 + 8} = 1 + 3 = 4 \text{ sec}$$

Ratio of total time taken to hit the ground to time taken to come back to same elevation

$$= \frac{4}{2} = 2 : 1$$

5. $u = 20\sqrt{3}$, $\theta = 60^\circ$

Angle made by velocity after time 't' with horizontal

$$\tan \alpha = \frac{u \sin \theta - gt}{u \cos \theta}$$

$$\Rightarrow \tan 30^\circ = \frac{20\sqrt{3} \times \sin 60^\circ - 10 \times t}{20\sqrt{3} \times \cos 60^\circ}$$

$$\Rightarrow \frac{1}{\sqrt{3}} \times 20\sqrt{3} \times \frac{1}{2} = 20\sqrt{3} \times \frac{\sqrt{3}}{2} - 10t$$

$$\Rightarrow 10t = 30 - 10 = 20$$

$$\therefore t = 2 \text{ sec}$$

8. $u \cos \theta = \frac{u}{2} \Rightarrow \cos \theta = \frac{1}{2}$, $\therefore \theta = 60^\circ$

$$R = \frac{u^2}{g} \sin 2\theta = \frac{u^2}{g} \sin 120^\circ = \frac{\sqrt{3}}{2} \frac{u^2}{g}$$

9. $t_f = 3 \text{ sec}$; $R = 150 \text{ m}$

$$\frac{2u \sin \theta}{g} = 3 \Rightarrow u \sin \theta = 15$$

$$H = \frac{u^2 \sin^2 \theta}{2g} = \frac{15^2}{2 \times 10} = 11.25 \text{ m}$$

10. For collision

$$V_2 \sin 30^\circ = V_1$$

$$\frac{V_2}{V_1} = \frac{1}{\sin 30^\circ} = 2$$

12. $y_1 + y_2 = \frac{u^2 \sin^2 \theta}{2g} + \frac{u^2 \cos^2 \theta}{2g} = \frac{u^2}{2g}$ $\begin{cases} \theta + \alpha = 90^\circ \\ \therefore \alpha = 90^\circ - \theta \end{cases}$

13. Angle made by velocity with horizontal after time 't'

$$\tan \alpha = \frac{u \sin \theta - gt}{u \cos \theta}$$

At $t = 2 \text{ sec}$, $\alpha = 0^\circ$, $\Rightarrow u \sin \theta = 2g = 20$

At $t = 1 \text{ sec}$, $\alpha = 45^\circ$, $u \cos \theta = (2g - g) = 10$

$$\therefore u = \sqrt{20^2 + 10^2} = 10\sqrt{5} \text{ m/sec}$$

15. Same as (13)

$$\tan \theta = 2; \quad \theta = \tan^{-1}(2)$$

17. $y = \sqrt{3}x - \frac{1}{2}gx^2$

eqⁿ of trajectory; $y = \tan \theta x - \frac{gx^2}{2u^2 \cos^2 \theta}$

on comparing we get,

$$\tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$$

$$\text{and } u^2 \cos^2 \theta = 1$$

$$\Rightarrow u^2 \times \frac{1}{4} = 1$$

$$\therefore u = 2 \text{ m/sec}$$

Projectile Motion

$$18. H = \frac{1}{4} R \Rightarrow \frac{H}{R} = \frac{\tan \theta}{4}$$

$$\Rightarrow \tan \theta = 1; \theta = 45^\circ$$

$$k_{\min} = k \cos^2 \theta = \frac{k}{2} = \frac{p^2}{4m} \quad \left[k = \frac{p^2}{2m} \right]$$

$$24. R = (2 + 4)m = 6m$$

$$x = 2m$$

$$\text{height of tower (y)} = ?$$

$$\text{town is at } x = 2m$$

$$y = \tan \theta \left(x - \frac{x^2}{R} \right) = 1 \left(2 - \frac{4}{6} \right)$$

$$28. \text{For a given velocity } v \text{ of projectile } y = \frac{4}{3} m$$

$$26. R = \frac{u^2}{g} \sin 2\theta \Rightarrow \frac{u^2}{g} = \frac{90^\circ}{\sin 120^\circ} = \frac{90 \times 2}{\sqrt{3}}$$

$$\text{Again, } R' = \frac{u^2}{g} \sin 2 \times 30^\circ = \frac{u^2}{g} \times \frac{\sqrt{3}}{2} = 90m$$

$$29. \text{Angle with horizontal}$$

$$\tan \alpha = \frac{gt}{u} = \frac{10 \times 10}{500} = \frac{1}{5}$$

$$\alpha = \tan^{-1} \left(\frac{1}{5} \right)$$
